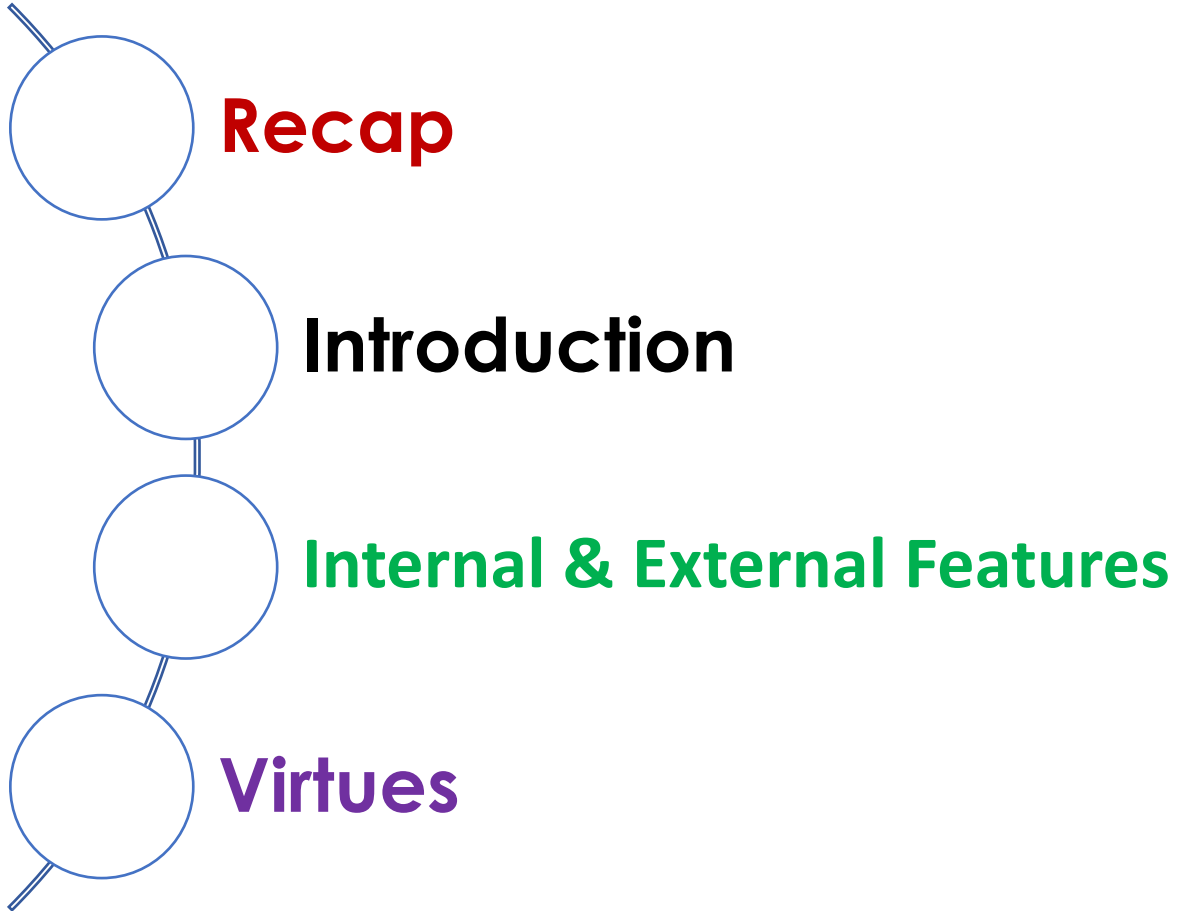


Internal & External Virtues



Alfred Wegener (1880 – 1930)
Meteorologist and geophysicist
Proposed the theory of continental drift

The Plan!



RECAP



INTRODUCTION

- Most people believe that science is a reliable way to acquire knowledge (justified true beliefs)
- But why should we believe what science says about the world?
 - This is a question of ***justification of scientific knowledge***
 - This question asks not so much what science claims about the world but more how science **proves** what it does claim and why it **gives us reason** to believe that these claims are true.



INTRODUCTION

- Science is largely about *unobservable* (theoretical) entities (“electrons”, “black holes”, etc).
 - For instance, the sentence “iron turns red when it is heated” can be verified because we can directly observe the iron, the redness, the heat, etc. The phenomena are *observable*.
 - But how can we verify the sentence “Helium atoms contain two electrons”?
 - We cannot directly observe these things!

INTRODUCTION

- Is the success of scientific theories a good indicator of their truth?



Reflect and discuss:

1. *“Science gains credibility from its phenomenal success. Airplanes fly. Smallpox has been done away with. Humble uranium gets turned into electricity, which in turn keeps the Bud Light cold. This record of success may be enough to give us responsible warrant to believe that science, by and large, gets at the truth about the world ...”*

Can the phenomenal success of science be considered sufficient in the justification of scientific claims? Why?



INTRODUCTION

What counts as success of a theory that is about things that can never be directly observed or experienced?

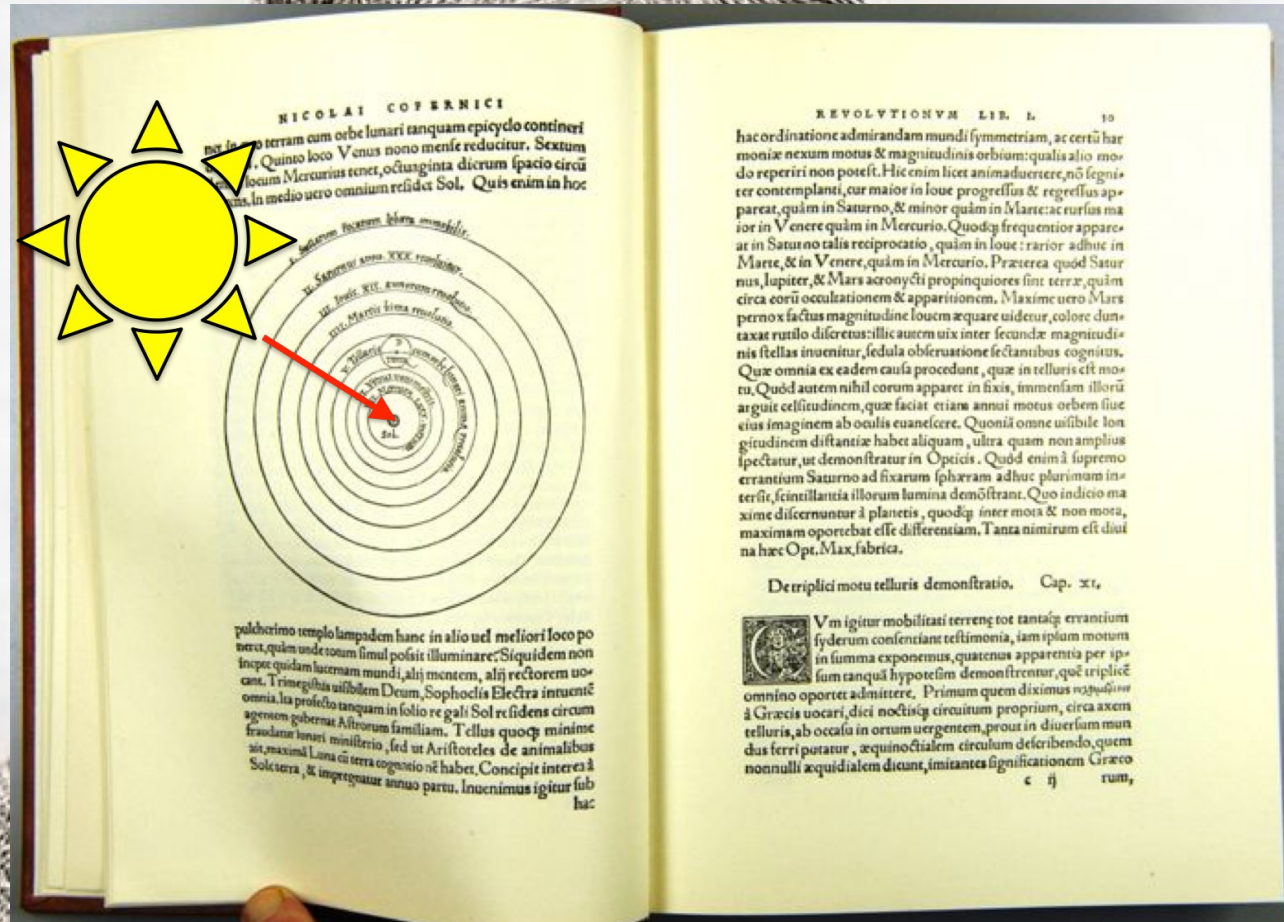
- **(1)** The criteria and measures of success are **proclaimed** by scientists themselves. Science, in this sense, is a largely **self-regulating** business.

Last time!

INTRODUCTION

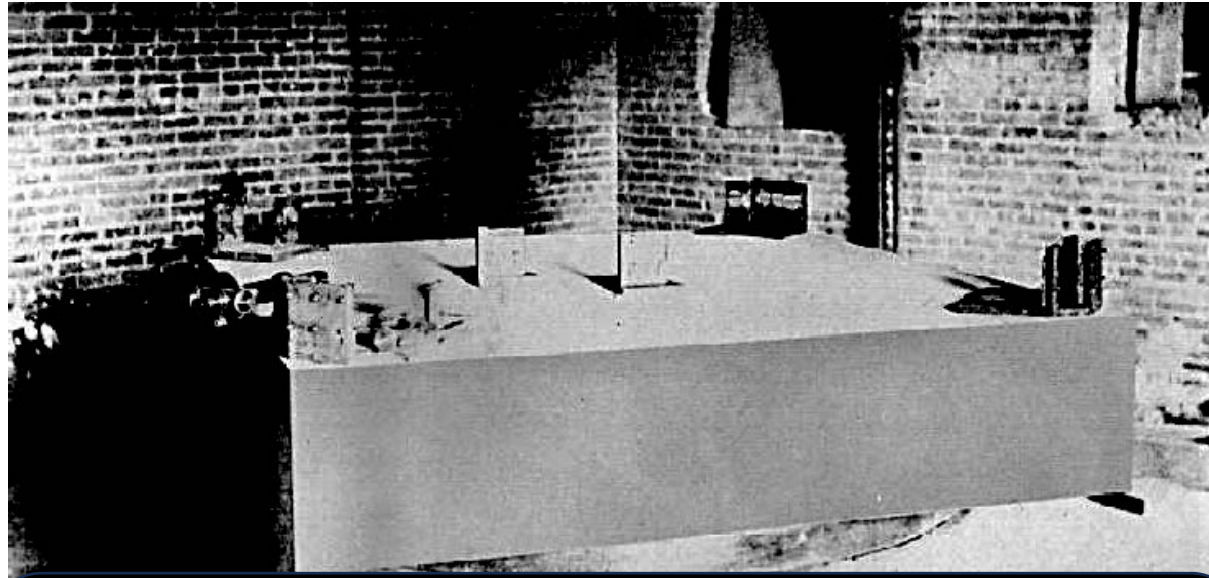
(2) Scientific theories change over time.

Nicolaus Copernicus
(1473–1543)



INTRODUCTION

(3) To say that something is done scientifically is not to say simply that it is done with success.



The Michelson-Morley experiment (1887) is a famous experiment in physics that tried and failed to detect the motion of the Earth through the 'aether'.



INTRODUCTION

- What is this methodological structure (or scientific method), and what about it gives us reason to think it produces truths?

WHAT IS SCIENCE?

Last time!

- The scientific method cannot be identified with a particular set of experimental or observational procedures
 - Methods vary:
 - In physics, biology, and chemistry, controlled experiments to test hypotheses are often possible
 - Epidemiology, medicine, and psychology often employ “natural” experiments
 - In astronomy and geology, hypotheses must usually be tested through observation

There are nonetheless some common features that are significant to justification of scientific claims



THEORIES

- The formulation of *theories* is the primary product of the scientific enterprise
 - **Theory (Def.)**: Any claim about the external world, general or specific, well justified or speculative, is a theory. Theories are the elements of expansive knowledge in that they always make claims to more than is told by the evidence.



THEORIES

- Science aims
 - to understand the composition and underlying causes of the **manifest** phenomena and thereby make sense of the phenomenal world we inhabit.
 - to offer **theories** that explain why the phenomena occur and what causes them.
- Science
 - is 'speaking about things **beyond** the evidence'
 - brings on a degree of uncertainty
- Theories are the primary subjects of **justification**



THEORIES

- 1) Claims that are only about unobservables without any direct impact on the phenomenal world
 - All bayrons (protons) are made of quark.
 - Helium atoms contain two electrons
- 2) Claims about the interactions between unobservables and observable.
 - Cloud Chamber

Both claims describe more than what is apparent, and in doing so they draw the question of justification.

HYPOTHESES & LAWS



- **Hypothesis (Def.):**
 - The particular theory that is being tested in a certain case is the hypothesis in that case.
 - This term is used to identify a theory (or theories) that is in need of testing.
 - Thus a hypothetical claim is made without much commitment to its truth. It is a suggestion in want of confirmation.

HYPOTHESES & LAWS



- Theories come in degrees of generality.
- Theories, which generalize and describe kinds of things, are **laws**.

HYPOTHESES & LAWS



- **Law (Def.):**

- A claim that associates a general kind of thing with a particular behavior or property is a law.
- Laws identify natural kinds in the world and imply a causal connection between being that kind of thing and having the associated property or behavior.
- Claims about accidental associations and the circumstantial properties of such groups are not laws, even though they are generalizations.
 - ***Laws support prediction; accidental generalization do not.***

- To sum up:
 - The concepts of theory (a claim about events that are not apparent to experience), hypothesis (not well-tested), and law (a generalization about a natural kind of object) describe **different aspects** of scientific claims and are not exclusive of each other.
 - There is no graduation in terms of ranks between them (first being hypothetical, then being a theory, and finally being a law).
 - The general theory of relativity is an example of a hypothetical law.
 - No theory reaches the level of certainty, of being beyond doubt and immune from change.
 - ***always a need for the process of justification***

INTRODUCTION

INTRODUCTION

Plenum:

1. “When we claim to have done something scientifically, we speak with the authority of a truth-generating enterprise.”

Many societies and organizations claim to venerate truth; can you think of any famous mottos that make truth their subject?

INTRODUCTION



we claim
in the authori

**Many societies and
think of any famous**



Phi Sigma
Biological Sciences Honor Society



we speak

**venen...th; can you
their subject?**

"Truth shall spring out of the
Earth"

INTRODUCTION



"Nullius in verba" (Latin for "no one's words" or "take nobody's word for it"; motto of the *Royal Society of London*)

- Science aims for justified-true beliefs (as ***likely to be true***)
- We said earlier that the formulation of theories is primary product of science.
- What distinguishes a good (strong) from a bad (weak) theory?

INTRODUCTION

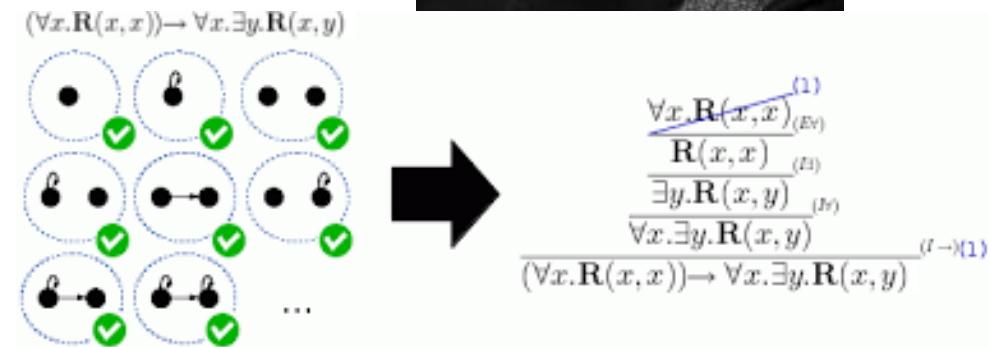
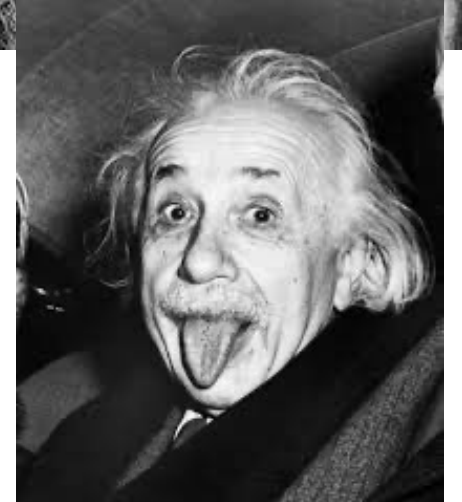
- A **strong theory** is one that we have good reason to believe is true, or at least likely to be true.
- A **weak theory** is one that, while perhaps attempting to explain a phenomenon, is highly unlikely to be true.
 - e.g., Miasma Theory: An outdated theory that assumed that diseases, such as the cholera or the plague, were caused by a noxious form of “bad air”.

INTRODUCTION

- A **strong theory** is one that we have good reason to believe is true, or at least likely to be true.
- A **weak theory** is one that, while perhaps attempting to explain a phenomenon, is highly unlikely to be true.
 - Theories have **features** that are ***indicative of their truth***,
 - The task of ***justification*** is to ***identify these features*** and use them to guide our choices as to which theory to believe.

INTRODUCTION

- Theories can have many different features:
 - when and where it was discovered
 - how it was discovered
 - by whom it was discovered
 - degree of generality and hypothetical status of a scientific claim
 - logical consistency in its form
 - pragmatic features (e.g., money-makers)
 - ...

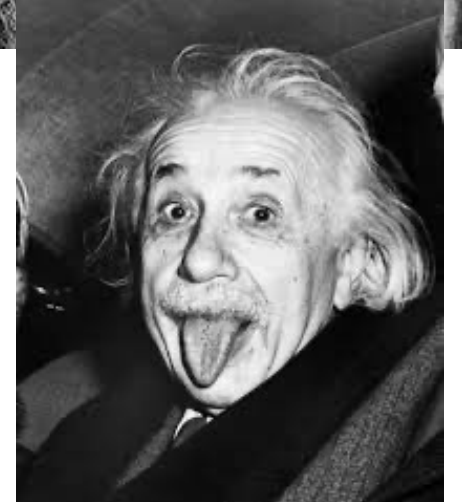


INTRODUCTION

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...

Some of these features are **irrelevant** to the justification of scientific claims and to doing science.



$$(\forall x.R(x,x)) \rightarrow \forall x.\exists y.R(x,y)$$



$$\frac{\frac{\frac{\forall x.R(x,x)}{(1)} \quad R(x,x)}{\exists y.R(x,y)} \quad (I)}{\forall x.\exists y.R(x,y)} \quad (I-)(1)$$

INTRODUCTION

- Features, relevant to the justification of scientific theories, must meet **two criteria**:
- They are
 - 1. truth-conducive**: They must be *relevant* to the likelihood of the theory being true.
 - Examples:
 - The feature that you learned about a theory on Tuesday is unimportant because it is neutral to the theory being true or false.
 - The feature of describing goblin maintenance men to keep the world in good working order might be an important feature that helps us to decide whether the theory is likely to be true or not.
 - 2. accessible**: A feature we can evaluate.
 - Information must be accessible to us if it is to be of any use.

INTERNAL & EXTERNAL FEATURES

INTERNAL & EXTERNAL FEATURES



Reflect & Discuss:

2. “Initially, then, features of theories can be grouped into those that are internal and those that are external.”

- **Explain the distinction between these two types of features.**
- **Does it relate in anyway to the text on “Empiricism & Rationalism”?**

INTERNAL & EXTERNAL FEATURES

- **Internal Features:**

- can be evaluated without having to observe the world
 - e.g., by studying books, that is, by looking at the structure of the theory and its conceptual relation to other theories.
- *One example of an internal feature is:* internal logical consistency (free of contradictions)
 - Example: Galileo's refutation of Aristotle's theory of motion.
 - Galileo showed that the theory implied contradictory things

INTERNAL & EXTERNAL FEATURES

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- **External Features:**

- are relevant to the theory's relation to the world, and they require observation for their evaluation
 - e.g., by performing an experiment and observing the outcome

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- ***Good science needs both internal and external features***

INTERNAL & EXTERNAL FEATURES

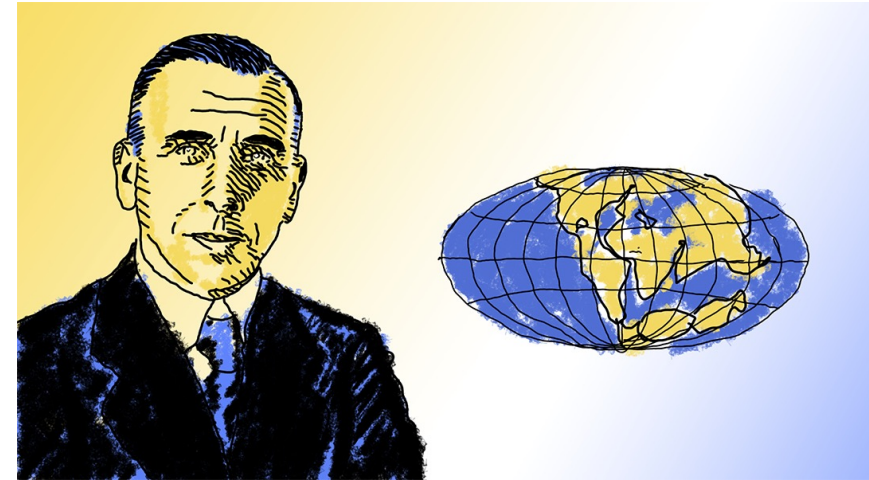
- Another potential expression of the relevant distinction between kinds of features focuses on the difference between **features of coherence** and **features of correspondence**
 - These concepts are often used by philosophers to describe features of beliefs we have, whether or not the beliefs deal with issues of science.

INTERNAL & EXTERNAL FEATURES



Reflect and discuss

3. Define *correspondence* and *coherence*
- Read the passage on your worksheet and make notes of any important points and definitions.



INTERNAL & EXTERNAL FEATURES

- **Coherence:**

- is descriptive of a *whole system of beliefs* and indicates that constituents fit together in some way.
- describes *relations* between one belief and other beliefs.
- presupposes consistency:
 - *Examples:*
 - The set of claims "The moon is full," "All dogs are vicious," "Smoking is an unhealthy activity," is a consistent set of claims (since no claim contradicts any other), but it is not coherent as a set.
 - "The moon is full," "The moon affects the tides," "The tide is higher than usual," is a consistent set of claims and has a high degree of coherence.

INTERNAL & EXTERNAL FEATURES

- **Coherence:**

- is descriptive of a *whole system of beliefs* and indicates that constituents fit together in some way.
- describes *relations* between one belief and other beliefs.
- presupposes consistency.

- **Correspondence:**

- describes the relations between one belief and the world.
 - Example: It's one thing for my belief that my little brother steals my cookies fits nicely with my belief that he is a poor student; it's quite another for my belief to correspond to the fact of the matter as to whether or not he is stealing the newspaper.

INTERNAL & EXTERNAL FEATURES

- For now, we will stick with internal and external features

VIRTUES

VIRTUES

- Scientists focus on the features of a theory that indicate that it is a good theory. These are *virtues*.
 - These are the important factors for comparisons of theories, that is, for deciding between competing accounts of the world. .
- Different aims of theorizing will work with different standards of goodness.
- Thus, we might ask:
 - *Good for what?*

VIRTUES

- In this context, we need to talk about the **epistemic** (true-conducive features) and **non-epistemic** virtues of scientific theories,
- which refer to the qualities or features that a theory might possess that influence its acceptability, usefulness, and overall value within scientific practice.

VIRTUES



Reflect and Discuss

4. In your pre-lecture reading, you have learned primarily about some *epistemic virtues* (truth-conducive) of scientific theories. What non-epistemic virtues were also mentioned?

Give examples of theories that possess these virtues.

VIRTUES

- **Non-Epistemic Virtues of Scientific Theories:**
 - These are qualities that contribute to a theory's usefulness, social relevance, or practical application but are ***not directly related*** to its ***truth*** or ***empirical adequacy*** (to accurately describe and predict observable phenomena).

VIRTUES

- **Non-Epistemic Virtues of Scientific Theories:**

- ***Some Examples:***

- *Pragmatic Virtue:* A theory might be valued for its practical utility or ability to solve a real-world problem (e.g., quantum mechanics has an immense practical value in technologies like semiconductors and GPS).
 - *Aesthetic Virtue:* A theory might be valued for its symmetry or beauty that makes it more aesthetically appealing to the eye (e.g., string theory or general relativity).
 - *Ethical or Social Virtue:* A theory might be valued for its ethical or social consequences (e.g., theories that promote environmental sustainability or social justice).

VIRTUES

- **Non-Epistemic Virtues of Scientific Theories**

- ***Some Examples:***

- *Pragmatic Virtue:* A theory might be valued for its practical utility or ability to solve a real-world problem. Quantum mechanics has an immense practical value in the development of technologies like semiconductors and GPS).

- *Aesthetic Virtue:* A theory might be valued for its symmetry or beauty that makes it appealing to the eye (e.g., string theory or general relativity).

- *Ethical or Social Virtue:* A theory might be valued for its ethical or social implications. For example, theories that promote environmental sustainability or social justice.

These kinds of features of theories are nice, but they are not what we are interested in ...

VIRTUES

- **Epistemic Virtues (truth-conducive virtues) of Scientific Theories:**
 - These are the features that make a scientific theory better at producing true, reliable, and justified beliefs.
 - In other words, they contribute to a theory's epistemic value (how well it helps us understand the world).

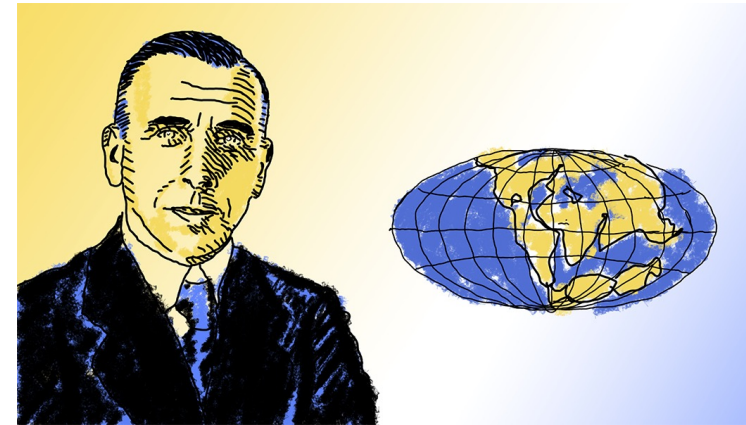
VIRTUES



Reflect and Discuss

5. Imagine being a philosopher who was a contemporary of Wegener. You proclaim:
"Within science, justification of a theory is the presentation of virtues."

How would you proceed in the evaluation of his theory?



VIRTUES

- **Epistemic Virtues (truth-conducive virtues) of Scientific Theories:**
 - **Internal Virtues:**
 - Internal features of theories that are truth-conducive
 - **External Virtues:**
 - External features that indicate that the theory is compatible with experimental results, that is, that the theory is accountable to observations.